

New Project - Big G Express

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<https://classroom.github.com/a/p1UvARqB>

Data can be downloaded from [here](#).

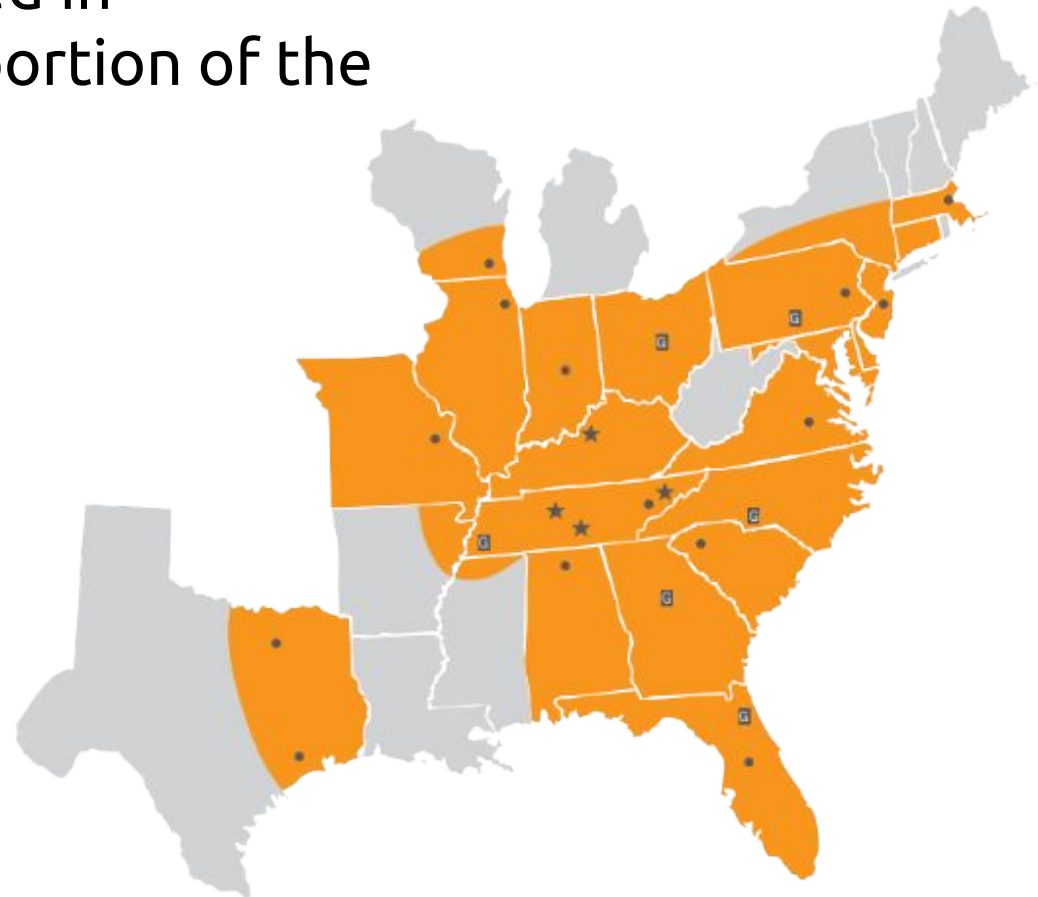
Do not push data to GitHub!!!

The logo for 'BIG G EXPRESS' is prominently displayed. 'BIG' is in a bold, black, sans-serif font with a thick orange outline. Below it, a large, stylized 'G' is rendered in orange with a black outline. To the right of the 'G', the word 'EXPRESS' is written in a bold, black, sans-serif font, slanted upwards to the right, with a thick orange outline.

Big G Express is a trucking company located in Shelbyville, TN and servicing the eastern portion of the United States.

G

- Memphis, Tennessee 
- Morristown, Tennessee 
- Nashville, Tennessee 
- Shelbyville, Tennessee 
- Dallas, Texas
- Houston, Texas
- Richmond, Virginia
- Milwaukee, Wisconsin



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Diesel emissions contain a number of problematic components:

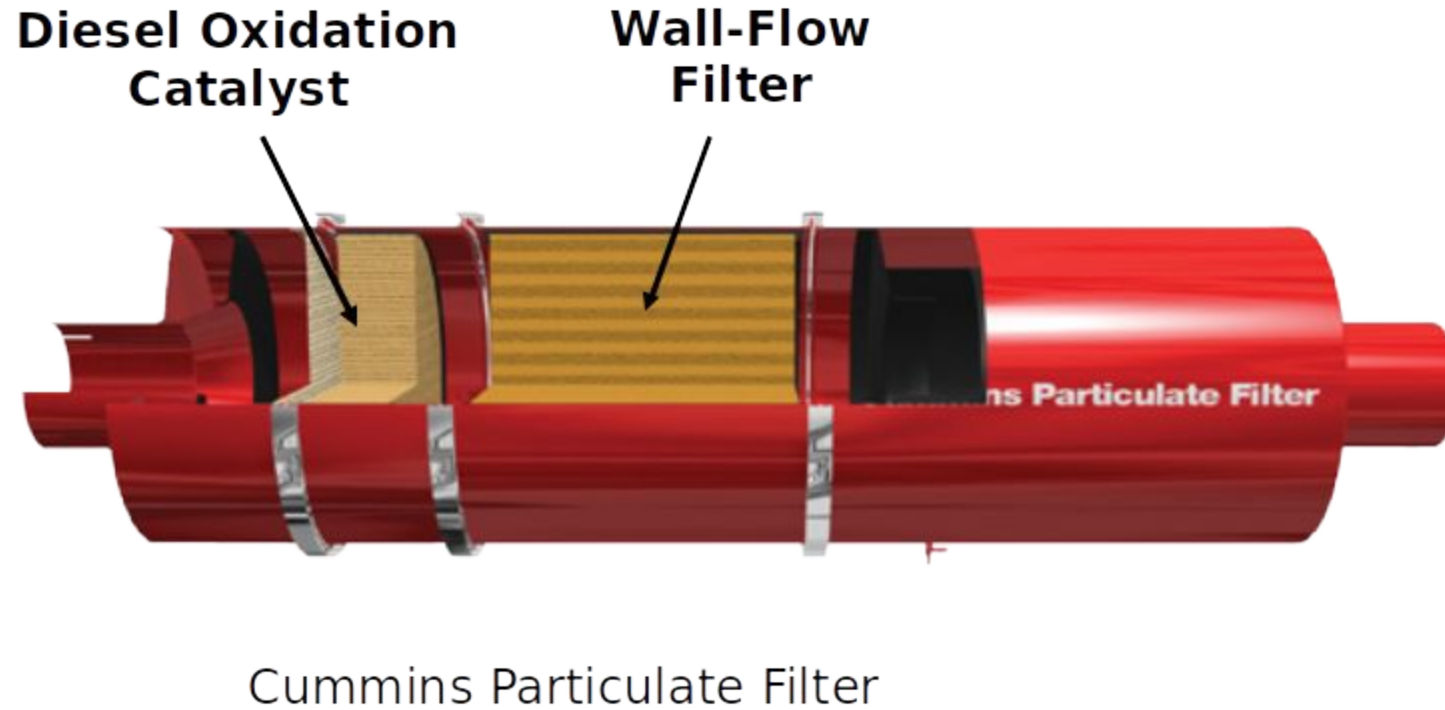
- Every gallon of diesel produces ~22.3 lbs of carbon emissions (total lifecycle)
- NOx (Nitrous Oxide), a precursor to ozone.
- Particulate matter (soot) – contains sulfates, ammonium, nitrates, some heavy metals, among other items.

New Project - Big G Express

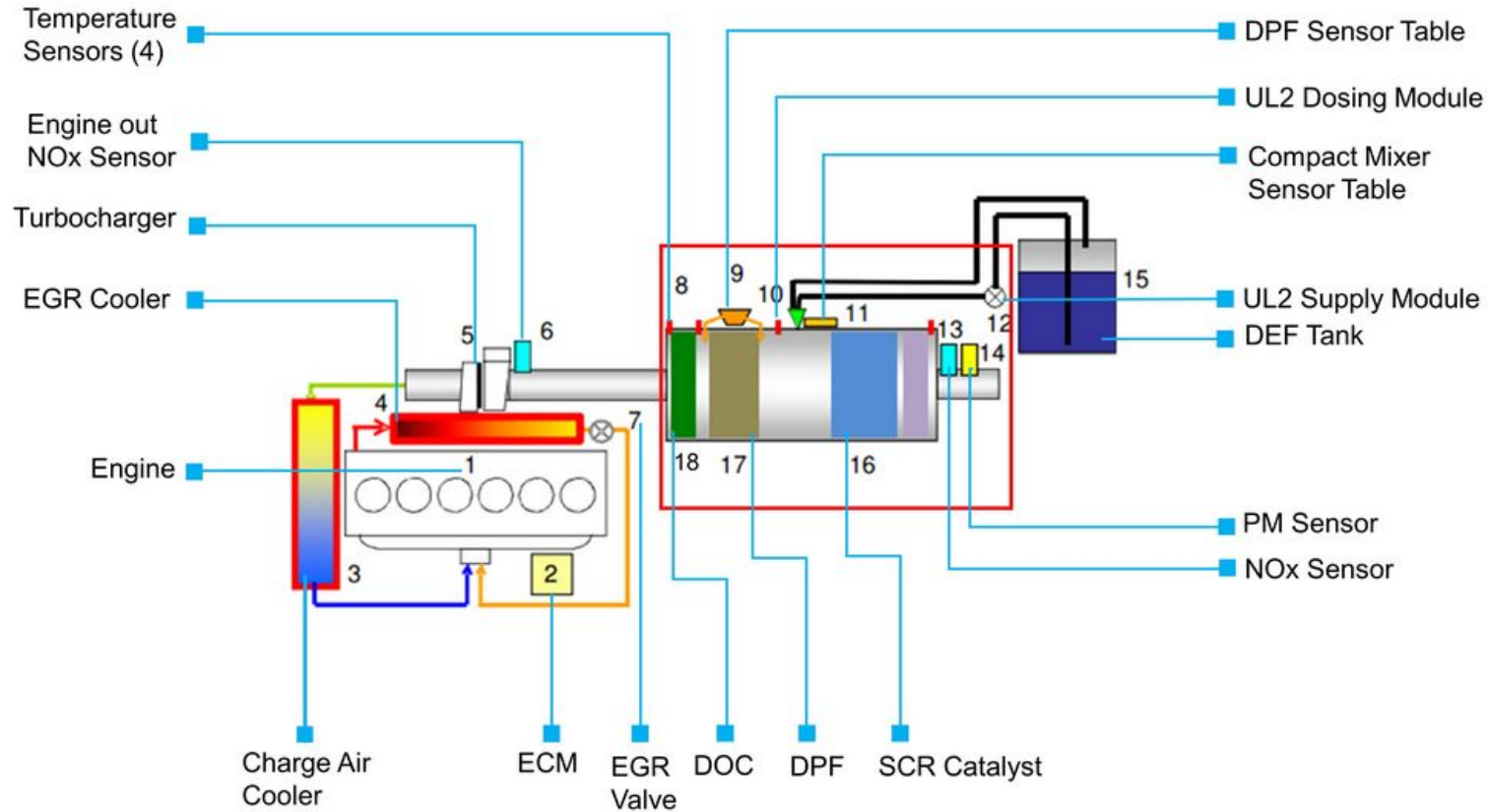
Industry Solutions:

- ULSD (Ultra-low Sulphur diesel fuel) was standardized nationwide in 2006. ULSD reduces the amount of NOX emissions, and reduces the hydrocarbon emissions to parity with gasoline.
- Diesel Particulate Filters – First used in cars as early as 1985, their use in heavy road tractors is near universal today. DPF's can reduce the amount of soot emitted by a vehicle by up to 95%.
- Idle Reduction – Most over the road tractors are equipped with Auxiliary Power Units, which provide power and driver comfort, so that the engine does not have to be idled. These are much more fuel efficient than idling the engine, and produce much lower emissions.

New Project - Big G Express



New Project - Big G Express



Cummins Aftertreatment System

See <https://www.cummins.com/components/aftertreatment/how-it-works> for more information.

New Project - Big G Express

There are a large number of sensors in the emissions system. The data you will be working with has records for when a fault in some component of the system is detected.

In certain cases, a **derate** occurs, which causes the engine output to be limited, and these are automatically triggered when the exhaust system detects a condition.

Your objective is to predict these with enough time to act on them and get the truck to a shop before the derate occurs and the truck needs to be towed.

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There are two main types of derate states:

- A 75% derate (SPN = 1569, FMI = 31) reduces engine torque by 25%.
 - This mode will require service to clear, but is generally manageable by routing the truck to the nearest maintenance facility.
- A idle level derate (SPN = 5246) will require a tow.
 - Not only will we miss delivery service on this load (which can sometimes drastically affect the customer's operations), but results in having to pay around \$4000 for towing, plus the cost of repairs.

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Your goal is to try and predict when a full derate (SPN = 5246) is imminent, but with enough time to get the truck off of the road. The more warning you can give, the better, but at least 2 hours is needed.

You need to balance being able to predict these full derate events ahead of time, but without too many false positives.

For purposes of evaluating your results, assume that the cost associated with a missed full derate is approximately \$4000 in towing and repairs, and the cost of a false positive prediction is about \$500 due to having the truck off the road and serviced unnecessarily.

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J1939Faults.csv

- Contains information about whether a fault light is going on (active = True) or turning off (active = False)
- Each fault has an spn + fmi
 - SPN = Suspect Parameter Number
 - FMI = Failure-Mode Indicator
- We will be most interested in partial derates (spn-fmi = 1569-31) and full derates (spn = 5246)

More information about the different fault codes is available in the Service Fault Codes spreadsheet.

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VehicleDiagnosticOnboardData.csv

- Contains vehicle diagnostic data from when a fault occurred
- Can be matched with the J1939 file using FaultId = RecordID
- **Warning:** Not everything is recorded for every fault code, so there are a large number of missing values for diagnostics. You can think about ways to still work with this data.

Service Fault Codes_1_0_0_167.xlsx

- Contains additional information about fault codes, eg. descriptions.
- Probably won't provide a lot of immediate benefit, so it is recommended that you prioritize this one last.
- **Warning:** For some spn/fmi combinations there are multiple records.